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Acute toxicity assessment of sodium fluoride to freshwater air breathing fish *Channa punctatus*: *In vivo* study

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Abstract- Fluorine being the most electronegative element in nature can react with many metals to form a variety of compounds, including sodium fluoride, which has been found to be toxic to a variety of organs in different animals like fishes, mice and humans. During the experiment, the freshwater fishes *Channa punctatus* were exposed separately to different doses of Sodium Fluoride (NaF) for 24 hrs, 48 hrs, 72 hrs and 96 hrs. At first, we determined median lethal concentration (LC_{50}) of NaF to fish using Probit analysis and confirmed it by pilot test. Changes in locomotory and behavioral attributes were observed carefully and monitored properly. The treated fishes showed restlessness and rapid random movements. Rapid opercular movements were also observed.

Key words: *Channa punctatus*, lethal concentration, pilot test, probit analysis, sodium fluoride.

INTRODUCTION

Today we are living in an environment where the purity of air, water and other components is in question. Changing human habits is aggravating the problem of environmental pollution and its contamination at a level where survival of plants and animals is becoming more and more difficult with passage of time. Along with some natural processes, like weathering of rocks, volcanoes, etc. anthropogenic activities are highly responsible for the same. Anthropogenic causes of environmental degradation are being ignored which are damaging the existing civilization directly or indirectly and causing a heavy loss of human health and economy on a large scale.

Toxicity assessment is a critical but significant process to detect as well as to predict the potential hazards of toxic substances. It is important in determining the level of sensitivity of organisms to different toxicants present in our ecosystem.[1] Acute toxicity test is performed in order to determine the lethal concentration of a toxic substance on test organisms which are exposed to a short span of time.[1]

After entering the body of fishes sodium fluoride hampers the activities of various enzymes disturbing many vital metabolic pathways.[2,3] Fluoride has been found to accumulate in the body of fish affecting its length and weight.[2,4,5] Fluoride toxicity has been correlated with several histopathological and biochemical alterations in several fishes, including *Heteropneustes fossilis*, *Labeo rohita*, etc.[6-8] Its genotoxic effect has been studies in *Clarias batrachus*.[9]

Fluoride moves from water to humans through intoxicated fishes, which are regular component of diet of

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large section of Indian as well as global population.[2] When such fishes are consumed regularly, fluoride accumulates in the body.[2] Its toxicity has also been closely linked to alterations in several vital cellular processes like cell division, apoptosis as well as their signaling pathways.[3] Many studies have confirmed the effects of fluoride toxicity in terms of adverse impacts on teeth and skeletal bones, causing several serious disorders.[10] Not only that its adverse effects have also been observed in many cells and soft tissues, like erythrocytes, thymocytes, liver, kidney and even brain in animals like rats and pigs.[11-15]

MATERIALS & METHODS

Healthy fishes, *Channa punctatus*, of weight 20-25 grams and length 10-15 cm were collected from Bazar Samiti, Patna, in July, 2021 and were brought to the Aquatic Toxicology Laboratory, Department of Zoology, Patna University, Patna. They were given prophylactic treatment of 0.01% $KMnO_4$ and after that actively moving fishes were transferred to different aquaria at a rate of 10 in each aquarium having a capacity of 90 litres containing well aerated and dechlorinated water. They were acclimatized for 15 days during which water was changed every 24 hours and they were given food *ad libitum* at 2-3% of body weight per day.

During the experiment, we used sodium fluoride of Merck Pvt. Ltd. We prepared test solution using deionised water. Stock solution of sodium fluoride was prepared as per the standard procedure of laboratory. The standard method of APHA (2005) was followed to perform the acute toxicity test. During this test period fishes were not provided any food since it could enhance the rate of various metabolic processes, which might have affected the level and result of toxicity of the test solution. Then we added different concentrations of sodium fluoride in different aquaria to conduct the test. Specific doses were determined by using trial and error method. The solution became turbid and the clarity of water reduced. The water in the aquaria was changed at the end of every 24 hours and the required level of test solution was monitored regularly during the experiment. The fishes were carefully observed and monitored for any change in behavioral and other responses. Dead fishes were removed immediately from the aquaria and their numbers were recorded carefully. After completion of the experiment the number of surviving fishes in each aquarium was counted and the observed behavioral and physiological responses were properly maintained. Using probit regression analysis (Finney, 1971) LC_{50} was calculated. Concentrations used were transformed logarithmically to obtain regression lines. Calculation of slope function(S), Upper and Lower confidential limit (UCL and LCL) of the regression line along with chi square test was done using standard protocol of UNEP/FAO/IAEA (1987).

RESULTS & DISCUSSION

The present study highlights the proportional increase in the mortality of fishes with increasing concentration of sodium fluoride. A linear relationship has been obtained between logarithm of concentration of sodium fluoride and logarithm of fish mortality (Probit) for almost all the groups of treated fishes (Table 1).

The calculated values of LC_{50} of sodium fluoride for 24 hours, 48 hours, 72 hours and 96 hours were 1479.11 ppm, 1047.13 ppm, 870.96 ppm and 851.14 ppm respectively by probit regression method. The regression equation mentioned in the Table 1 highlights that with the increase in period of exposure the mortality of fishes increased. Behavioral and physiological responses of the fishes also varied during the exposure. At the beginning of the experiment, fishes exhibited random movements and were seen to be restless. Later, they showed abnormal movements and behavioural response by moving haphazardly i.e., irregularly changing their directions. The rate of movement of operculum also increased appearing that they were trying to engulf more and more air. With passage of time these changes in responses became occasional. These changes might have been due to the stress caused by toxicity of sodium fluoride which affects several signaling pathways in organisms.

Physio-chemical factors of water like temperature, alkalinity, hardness, pH, etc. have been reported to affect the extent of toxicity caused by sodium fluoride.[16] Fluoride has been reported to have many adverse effects in animals like reduced body weight and size, inhibition of some of the essential enzymes like those involved in synthesis of proteins and glycolysis.[17]

Higher concentrations of fluoride has been found in fishes and it has been mainly due to the presence of fluoride-containing minerals in various components of the lithosphere like rocks and soils as well as their dumping into water bodies like sea.[18]

A large number of people in many Indian states are living at the risk of higher fluoride level than permissible i.e., 1.5 mg/L in their drinking water. Poor people are at the risk of being affected by fluoride more than other groups.[19]

For *Clarias gariepinus*, 96 hour LC_{50} of sodium fluoride has been found to be 734.51 mg/L.[20] In case of *Oncorhynchus mykiss*, 96 hour LC_{50} was earlier found to be 200 mg/L,[20] for *Salmo trutta*, it has been found to be 164.5 mg/L,[21] for *Gasterosteus aculeatus*, it has been 460 mg/L,[21] for *Pimephakles promelas*, it has been 315 mg/L,[22] for *Puntius sophore*, it has been found to be 126.12 mg/L[23] and for Amur carp (*Cyprinus carpio haematopterus*) it has been found to be 986.28 mg/L.[24]

Table 1. Showing various lethal concentrations of sodium fluoride to *Channa punctatus* at different exposure level

Exposure period (Hrs)	LC_{20} Value (ppm)	LC_{50} Value (ppm)	LC_{90} Value (ppm)	Regression equation Y=mX+C	95% confidence limit		P-value	Chi-square value	
					Lower limit	Upper limit		Calculated value	Table value
24	1318.26	1479.11	1778.28	16.87X-48.49	-184.44	87.45	0.253	2.722	7.815
48	1023.29	1047.13	1148.15	36.38X-105.20	-236.18	25.78	0.071	7.873	7.815
72	812.83	870.96	954.99	29.04X-80.32	-227.97	67.32	0.128	1.478	7.815
96	794.33	851.14	933.25	28.66X-78.99	-203.68	45.69	0.099	1.463	7.815

N = 10 for each group; chi square table value taken at 3df and $p<0.05$

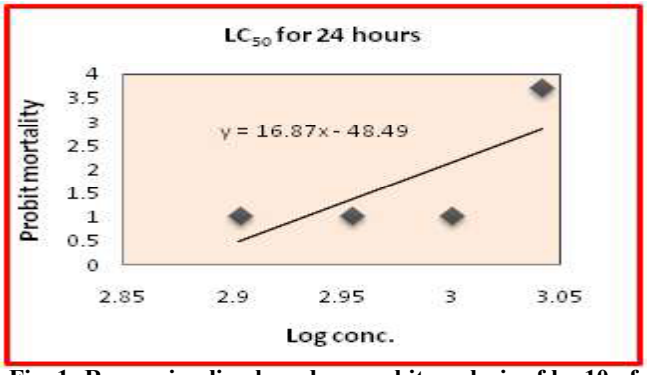


Fig. 1- Regression line based on probit analysis of log10 of concentration of sodium fluoride vs probit of percent mortality of *Channa punctatus* at 24 hours

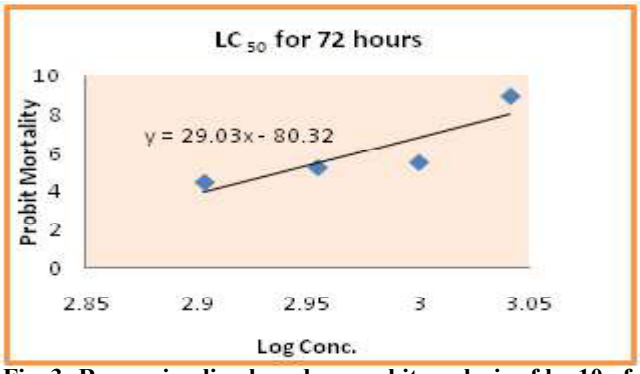


Fig. 3- Regression line based on probit analysis of log10 of concentration of sodium fluoride vs probit of percent mortality of *Channa punctatus* at 72 hours

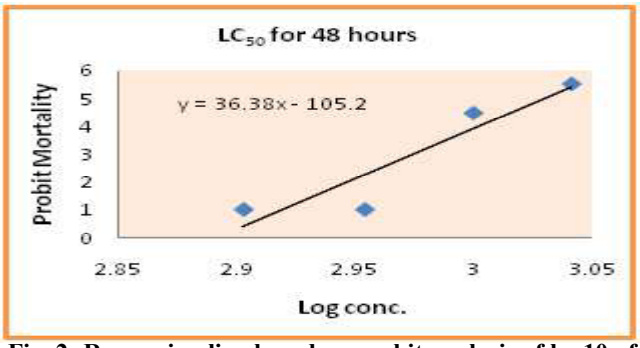


Fig. 2- Regression line based on probit analysis of log10 of concentration of sodium fluoride vs probit of percent mortality of *Channa punctatus* at 48 hours

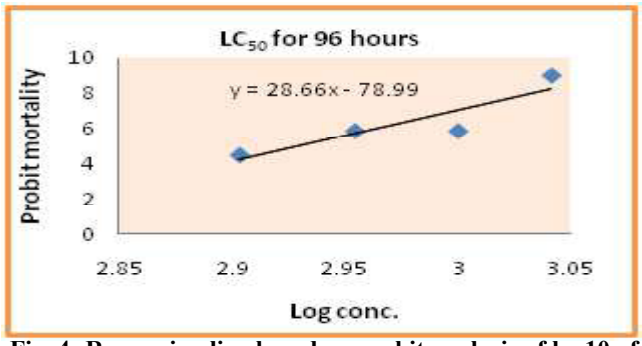


Fig. 4- Regression line based on probit analysis of log10 of concentration of sodium fluoride vs probit of percent mortality of *Channa punctatus* at 96 hours

CONCLUSION

The fish *Channa punctatus* were exposed to different concentrations of sodium fluoride and the LC_{50} values were calculated at 24 hours, 48 hours, 72 hours and 96 hours. We found here high values of LC_{50} which might have been due to various factors. Further studies are needed to understand the keyfactors that are playing vital role in variations in the LC_{50} values in *C. punctatus*.

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CONFLICT OF INTEREST

Authors declare no conflict of interest.

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